# Ellipse Perimeter: Ramanujan Approximation and N-Series Geometric Acceleration

A Chord-Cost and Sinc-Correction Comparison

C. Wayne Baker

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**Abstract**

We compare Ramanujan's second closed-form *ellipse-perimeter* approximation with inscribed equal-parameter polygons and their scale-extrapolated refinements. The three-scale geometric N-series operator cancels the $N^{-2}$ and $N^{-4}$ perimeter errors and converges at order $N^{-6}$ for each fixed nondegenerate ellipse. We distinguish its base resolution $N$ from its direct chord-evaluation cost $7N$. An exact chord factorization proved in a subsequent companion paper yields the single-polygon sinc-corrected approximation $P_N/\operatorname{sinc}(\pi/N)$, removing the entire algebraic chord-error factor. The remaining periodic midpoint-quadrature error decays exponentially with $N$ for fixed axis ratio $b/a > 0$. We reproduce the earlier high-precision comparisons, explain the near-universal normalized sixth-order coefficient, and give chord-count-aware tests of sinc correction. The three mechanisms are mathematically distinct: a closed-form eccentricity approximation, polynomial Richardson cancellation, and analytic chord normalization. No numerical superiority over established elliptic-integral algorithms is claimed.

## 1 Exact perimeter and Ramanujan's approximation

For semiaxes $a \geq b > 0$, the exact perimeter is

$$P = 4aE(e), \qquad e = \sqrt{1 - b^2/a^2}, \qquad E(e) = \int_0^{\pi/2} \sqrt{1 - e^2 \sin^2 t}\, dt. \tag{1}$$

Ramanujan's second ellipse-perimeter approximation is

$$R_2 = \pi(a+b)\left(1 + \frac{3h}{10 + \sqrt{4 - 3h}}\right), \qquad h = \left(\frac{a-b}{a+b}\right)^2. \tag{2}$$

This is an approximation to the ellipse perimeter, *not* a Ramanujan series for $\pi$. The signed error is positive at sufficiently small $h$: $P - R_2 = \pi(a+b)(3h^5/131072 + O(h^6))$. In the comparison tables we report *absolute* errors $|P - \widetilde{P}|$ in the units of $a$ and $b$.

## 2 Equal-parameter polygons and scale cancellation

Set $q(t) = (a\cos t, b\sin t)$ and $t_j = 2\pi j/N$. The perimeter $P_N$ is the sum of Euclidean chord lengths $|q(t_{j+1}) - q(t_j)|$, with $t_N = 2\pi$. For a fixed nondegenerate ellipse the symmetric discretization admits an even-power asymptotic expansion

$$P_N = P + C_2 N^{-2} + C_4 N^{-4} + C_6 N^{-6} + O(N^{-8}). \tag{3}$$

Two-scale cancellation gives $(4P_{2N} - P_N)/3 = P + O(N^{-4})$. The original three-scale N-series estimator is

$$T_N = \frac{P_N}{45} - \frac{4P_{2N}}{9} + \frac{64P_{4N}}{45} = P + O(N^{-6}). \tag{4}$$

Indeed its weights sum to 1 and its $N^{-2}$ and $N^{-4}$ moments vanish:

$$\frac{1}{45} - \frac{4}{9 \cdot 4} + \frac{64}{45 \cdot 16} = 0, \qquad \frac{1}{45} - \frac{4}{9 \cdot 16} + \frac{64}{45 \cdot 256} = 0.$$

**Cost accounting.** The three polygon sums use $N + 2N + 4N = 7N$ chord-length evaluations if computed separately; thus a row headed $N = 64$ represents a direct cost of 448 chords, not 64. Although sample points can sometimes be reused, the polygon chord lengths for different step sizes are not identical. We use chord counts as a transparent geometric operation proxy, *not* as a measured wall-clock or arithmetic-operation benchmark.

## 3 Exact sinc correction and the sixth-order coefficient

Let $v(t) = \sqrt{a^2 \sin^2 t + b^2 \cos^2 t}$ and define the midpoint sum

$$M_N = \frac{2\pi}{N} \sum_{j=0}^{N-1} v\left(\frac{2\pi(j + 1/2)}{N}\right).$$

The ellipse chord from $t_j$ to $t_{j+1}$ has length $2\sin(\pi/N)v(2\pi(j + 1/2)/N)$, exactly. Hence the factorization from the companion ellipse study [5] is

$$P_N = \operatorname{sinc}(\pi/N) M_N, \qquad \operatorname{sinc}(x) = \frac{\sin x}{x}. \tag{5}$$

Accordingly the single-polygon corrected value is

$$S_N = \frac{P_N}{\operatorname{sinc}(\pi/N)} = M_N. \tag{6}$$

The speed $v$ is analytic in each strip $|\operatorname{Im} z| < \rho_*$, where $\rho_* = \operatorname{artanh}(b/a)$ for $0 < b < a$. Standard periodic quadrature estimates imply $S_N - P = O(e^{-\rho N})$ for any fixed $0 < \rho < \rho_*$; for odd $N$ the $\pi$-periodicity gives the sharper bound $O(e^{-2\rho N})$. For a circle the midpoint rule is exact. These bounds are for a *fixed* ellipse; the constants are not uniform as $b/a \to 0$.

Since $M_N - P$ is exponentially small and

$$\operatorname{sinc}(\pi/N) = 1 - \frac{\pi^2}{6N^2} + \frac{\pi^4}{120N^4} - \frac{\pi^6}{5040N^6} + O(N^{-8}),$$

the algebraic sixth-order term of $T_N - P$ is

$$T_N - P = -\frac{P\pi^6}{5040 \cdot 64\, N^6} + O(N^{-8}) + O(e^{-\rho N}), \tag{7}$$

because $\frac{1}{45} - \frac{4}{9 \cdot 2^6} + \frac{64}{45 \cdot 4^6} = \frac{1}{64}$. Consequently, $N^6 |T_N - P|/P \to \pi^6/(5040 \cdot 64)$. This explains the nearly uniform small errors across ellipse shapes at sufficiently large $N$: they are proportional to the respective exact perimeters. At modest $N$ and high eccentricity the exponential midpoint contribution can dominate; a universal asymptotic prediction is then premature.

## 4 Numerical comparison and precision

The historical table and initial comparison were checked through AI-assisted high-precision recomputation; the newly added equal-cost $S_{448}$ entries were separately recomputed with 170-decimal-digit working precision using the formulas stated here. These are computational cross-checks, not independent human peer review. The original paper did not specify the precision used to generate its historical table; we therefore do *not* retroactively assert a particular original working precision. Reproduction should use arbitrary precision with at least 60 decimal digits, evaluating (1)–(6) without binary64 rounding of intermediate chord sums. Errors around $10^{-17}$ in perimeters of order five cannot reliably be resolved by subtracting two ordinary double-precision approximations. For

| $b$ ($a = 1$) | Base $N$ | Direct chords $7N$ | $\lvert P - R_2\rvert$ | $\lvert P - T_N\rvert$ |
|---|---|---|---|---|
| 0.90 | 256 | 1792 | $2.24 \times 10^{-17}$ | $6.32 \times 10^{-17}$ |
| 0.75 | 256 | 1792 | $4.61 \times 10^{-13}$ | $5.85 \times 10^{-17}$ |
| 0.50 | 256 | 1792 | $2.21 \times 10^{-9}$ | $5.13 \times 10^{-17}$ |
| 0.25 | 256 | 1792 | $1.07 \times 10^{-6}$ | $4.54 \times 10^{-17}$ |

Table 1: Absolute perimeter errors for Ramanujan's closed form and the original three-scale method; $N$ is the base resolution, not the chord cost.

$b = 0.50$, the original three-scale errors at base $N = 64, 128, 256$ are $2.10 \times 10^{-13}$, $3.28 \times 10^{-15}$, and $5.13 \times 10^{-17}$, approaching the factor-of-64 improvement per doubling predicted by (7). At $b = 0.25$, base $N = 32$ gives $1.81 \times 10^{-7}$, much larger than the sixth-order algebraic prediction because the quadrature remainder is still material. For $b = 0.50$, even $S_{64}$ is reported to attain an

| $b$ | $\lvert P - R_2\rvert$ | Base $N$ | $7N$ | $\lvert P - T_N\rvert$ | $\lvert P - S_{128}\rvert$ | $\lvert P - S_{448}\rvert$ |
|---|---|---|---|---|---|---|
| 0.50 | $2.2 \times 10^{-9}$ | 64 | 448 | $2.1 \times 10^{-13}$ | $1.4 \times 10^{-33}$ | $9.99 \times 10^{-111}$ |
| 0.25 | $1.1 \times 10^{-6}$ | 64 | 448 | $1.8 \times 10^{-11}$ | $2.2 \times 10^{-17}$ | $1.07 \times 10^{-53}$ |

Table 2: Comparison including $S_{128}$ (128 chords) and the exactly chord-count-matched $S_{448}$ (448 chords). Rounded absolute errors; the $S_{448}$ values were recomputed at 170-digit precision.

absolute error about $7.6 \times 10^{-18}$ from only 64 chords. Sinc correction is therefore preferable to the three-scale ladder for these particular numerical cases. Near the degenerate limit $b/a \to 0$, however, the analyticity strip collapses and $S_N$ need not yet be exponentially accurate at a given finite $N$. Moreover, (6) uses an analytic factor containing $\pi$; it is not a claim that straightedge-and-compass operations alone exactly correct the polygon.

## AI-assistance and numerical provenance

AI tools, including ChatGPT and Claude, assisted with editorial review, formula checks, and numerical comparisons. The earlier tables reflect reported high-precision AI-assisted checks, while the equal-cost $S_{448}$ entries were freshly calculated for this revision with 170-decimal-digit arithmetic. These computational checks do not constitute independent human refereeing. No new empirical runtime measurements are claimed.

## 5 Interpretation and conclusion

The original N-series experiment correctly demonstrates sixth-order Richardson cancellation and shows that sufficient polygonal refinement can exceed the accuracy of Ramanujan's single closed-form ellipse approximation for the tested axis ratios. Explicit cost accounting sharpens that finding: a three-scale estimator at base $N$ uses $7N$ chords, whereas a single sinc-corrected $N$-gon uses $N$ chords. The later exact factorization (5) explains why correcting the chord factor all at once can be far more accurate for an analytic ellipse speed. It also gives a precise interpretation of the original $N = 256$ table through the universal algebraic residual (7).

The three methods answer different questions. Ramanujan gives a striking fixed algebraic approximation along the eccentricity axis; N-series extrapolation cancels geometric resolution errors without relying on special ellipse factorization; sinc correction exploits exactly such a factorization and leaves periodic quadrature error. These results are not benchmarks against direct elliptic-integral evaluation, AGM methods, or optimized software implementations. Chord counts alone do not measure runtime. The relevant contribution is a transparent comparison of accuracy mechanisms and their geometric sampling costs.

## References


[1] S. Ramanujan, "Modular equations and approximations to $\pi$," *Quarterly Journal of Mathematics* 45 (1914), 350–372.

[2] J. M. Borwein and P. B. Borwein, *Pi and the AGM*, Wiley, 1987.

[3] P. F. Byrd and M. D. Friedman, *Handbook of Elliptic Integrals for Engineers and Physicists*, Springer, 1954.

[4] L. F. Richardson, "The approximate arithmetical solution by finite differences of physical problems," *Philosophical Transactions of the Royal Society A* 210 (1911), 307–357.

[5] C. W. Baker, *Exact Chord Factorization and Universal Asymptotics for Inscribed Ellipse Polygons*, companion manuscript, revised October 2026.

[6] L. N. Trefethen and J. A. C. Weideman, "The exponentially convergent trapezoidal rule," *SIAM Review* 56 (2014), 385–458.